KENYA FOOD PRICE PREDICTION

Using Machine Learning to Forecast Maize, Beans & Milk Prices Across Kenya

Group 6 · Moringa School · Data Science Programme

THE PROBLEM

Extreme Price Volatility

Essential food prices swing unpredictably — maize, beans, and flour fluctuate sharply month-to-month across Kenya's 27 tracked counties.

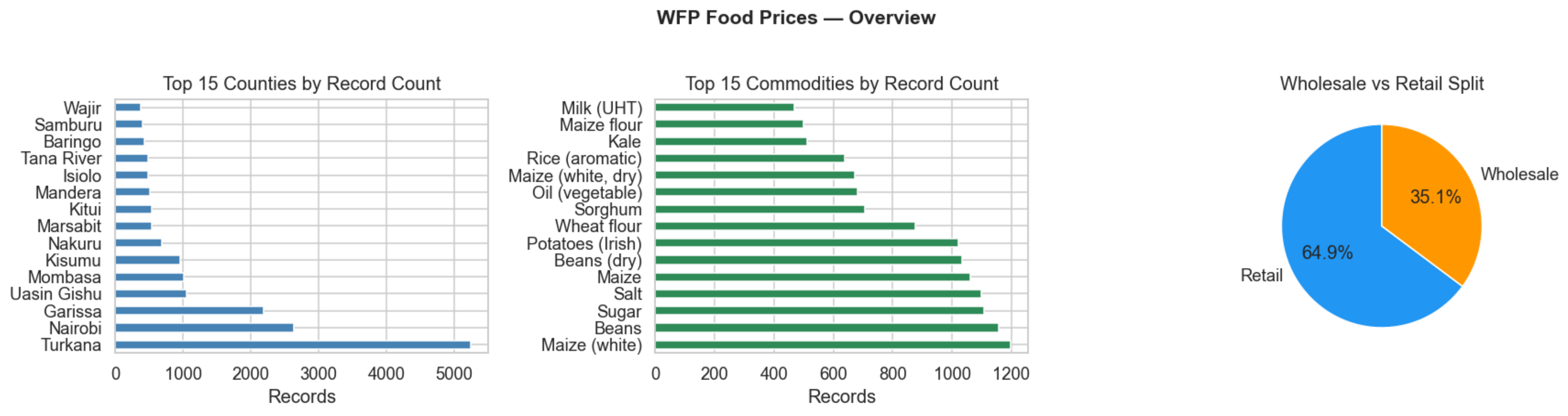
Farmers Left Guessing

Farmers in Uasin Gishu and other counties cannot plan planting cycles or decide when to sell without reliable price forecasts.

Household Budgets at Risk

Consumers face surprise price spikes that disrupt family food budgets, especially for the most vulnerable households.

THE DATA WE USED



19,005

Price records

27

Counties covered

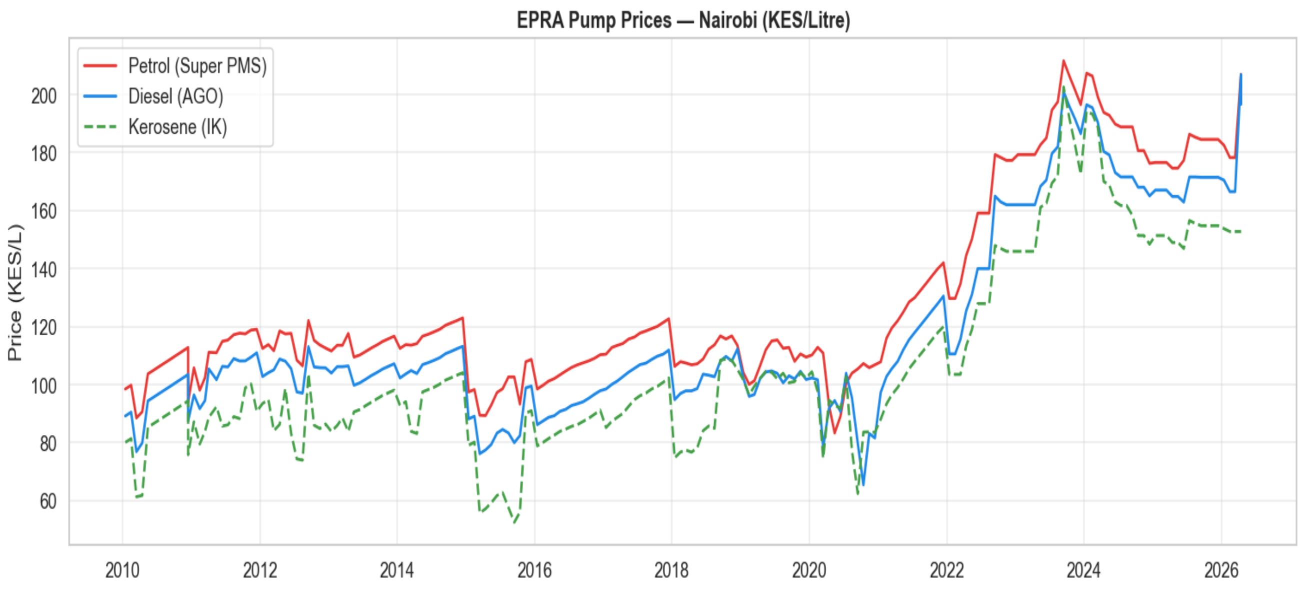
51

Commodities tracked

2006–2026

Date range

HOW FUEL COSTS AFFECT FOOD PRICES

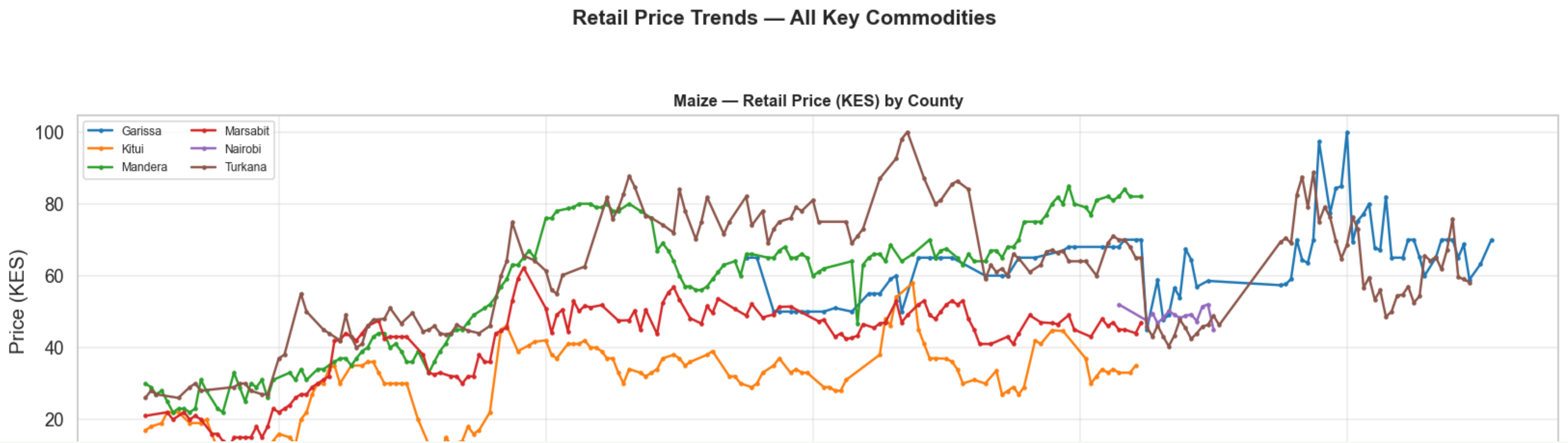


Key Findings

- Diesel prices surged from ~KES 100 (2021) to over KES 200 (2023–24) — a near doubling.
- All food commodities show positive correlation with diesel — transport costs affect every staple.
- 2020 dip aligns with COVID-era demand collapse; sharp recovery followed.

Higher fuel → higher transport cost → higher food prices at market (3–6 month lag)

HOW MAIZE PRICES HAVE MOVED (2006–2026)



KES 49.7

Mean price (all history)

KES 6–100

Full price range observed

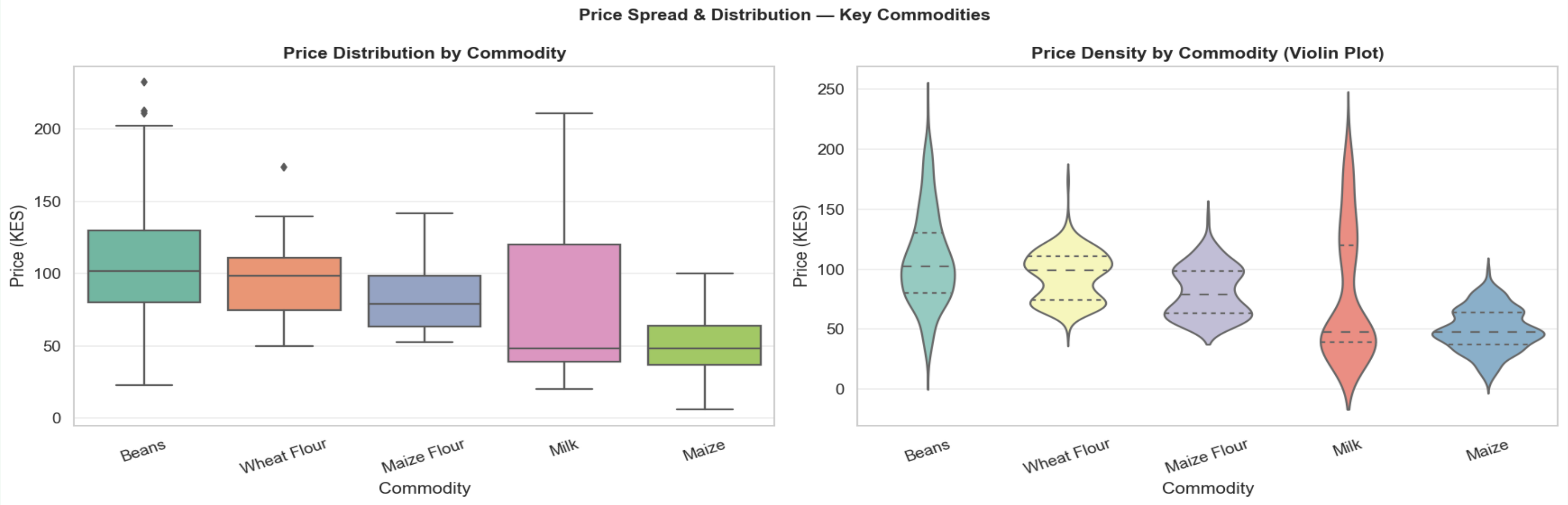
KES 17.8

Std deviation

Turkana

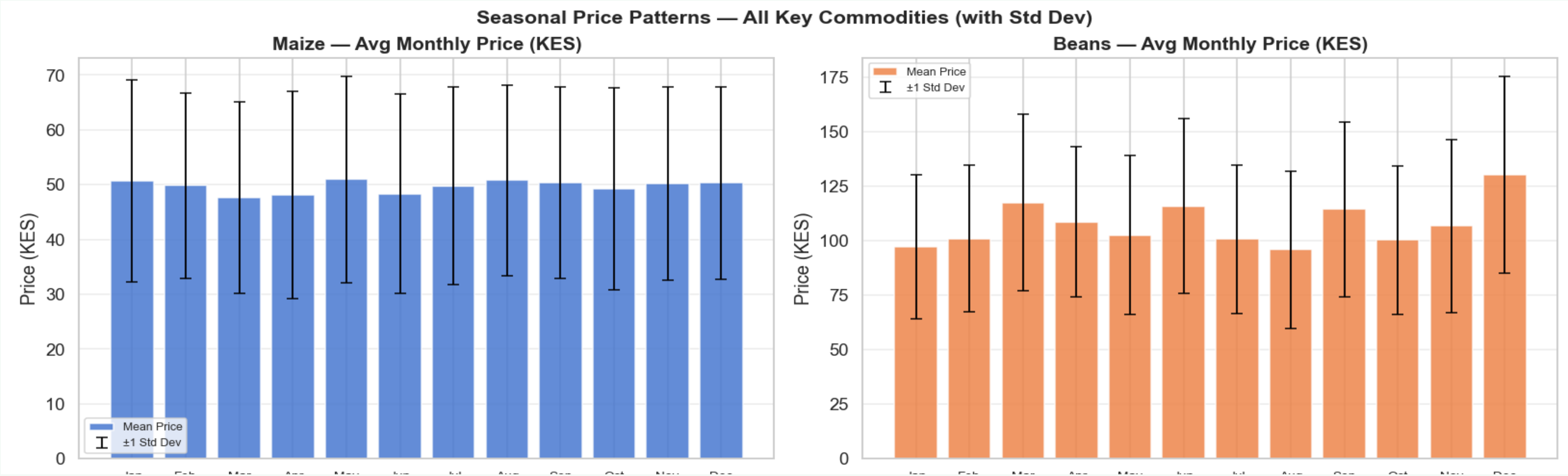
Highest-priced county

PRICE SPREAD ACROSS ALL COMMODITIES



Beans (KES 108 avg) are the most expensive staple · Maize (KES 50) is the most affordable · Milk shows the widest county-to-county gap

SEASONAL PRICE PATTERNS

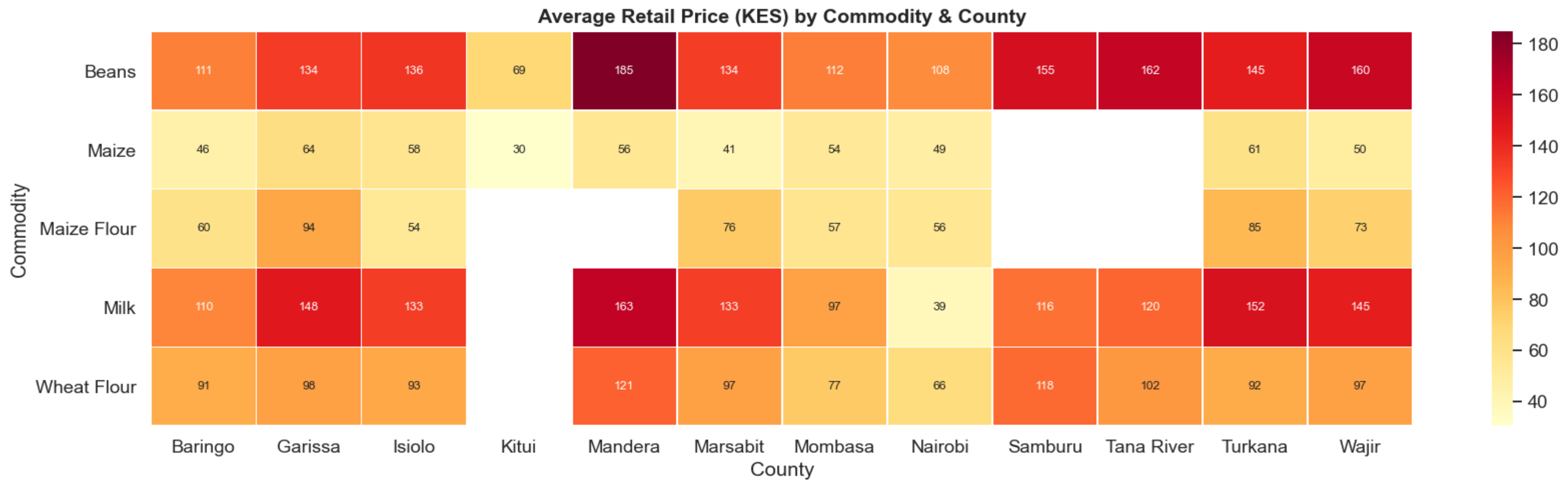


Maize: relatively stable year-round (KES 47–51). Slight lean-season peaks Jan–May before harvest.

Beans: March and December are the most expensive months — coinciding with demand peaks.

💡 These seasonal patterns are captured by the model's month_sin and month_cos features.

PRICES DIFFER WIDELY BY COUNTY



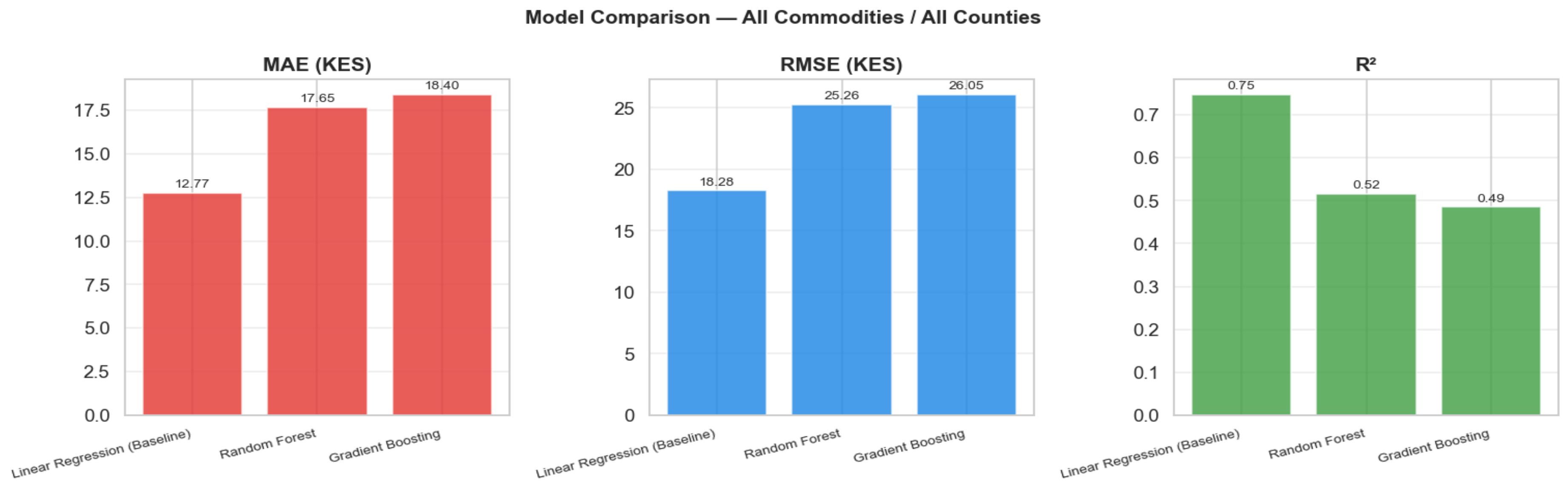
- Mandera has the highest Beans price (KES 185 avg) vs Kitui (KES 69) — a 2.7× gap for the same commodity.
- Nairobi is consistently cheaper for most staples due to better supply chains and market competition.
- Remote/arid counties (Mandera, Wajir, Marsabit) pay 30–60% more than urban markets.

RAINFALL AND FOOD PRICES

How Rainfall Affects Prices

- Price spikes in Maize and Beans follow dry periods with a 3–6 month lag — the time from planting failure to market shortage.
- The model includes 1-month, 3-month and 6-month lagged rainfall features to capture this delayed effect.

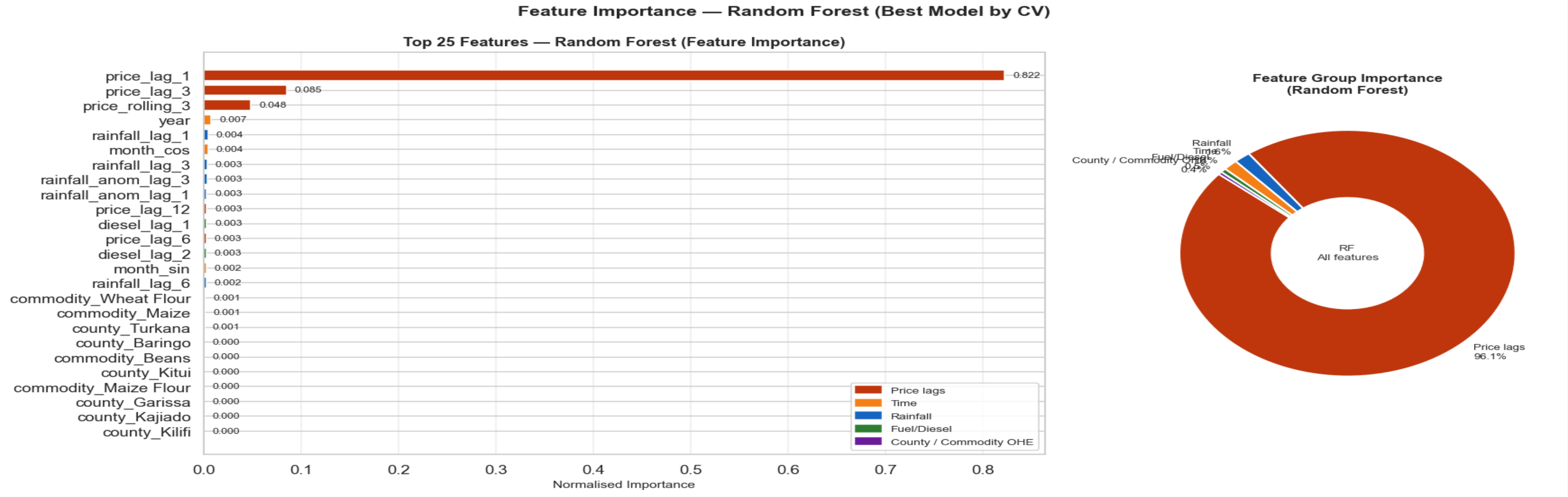
THREE MODELS COMPARED



🏆 **Winner (overall baseline): Linear Regression · MAE = 12.77 KES · R^2 = 0.75**

Per-commodity training (Phase 2) uses the best model for each individual commodity.

WHAT DRIVES THE PREDICTIONS?



Past Prices (96.1%)

price_lag_1 alone scores 82.2% — recent prices are by far the strongest signal.

Time / Season (0.7%)

year and month cyclical features capture inflation trends and harvest cycles.

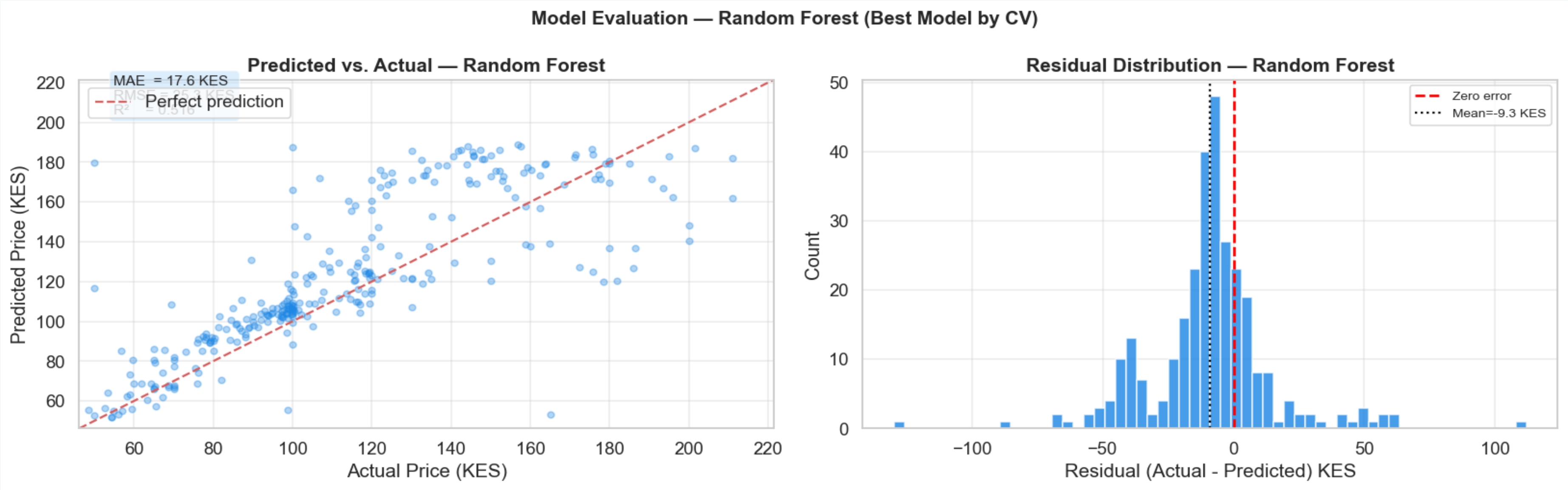
Rainfall (0.5%)

1-month and 3-month lagged rainfall features — modest but meaningful.

Fuel / Diesel (0.3%)

diesel_lag_1 and diesel_lag_2 — delayed pass-through from transport costs.

HOW WELL DOES THE MODEL PREDICT?



17.6 KES

Average Error (MAE)

Random Forest, all commodities combined

−9.3 KES

Mean Residual

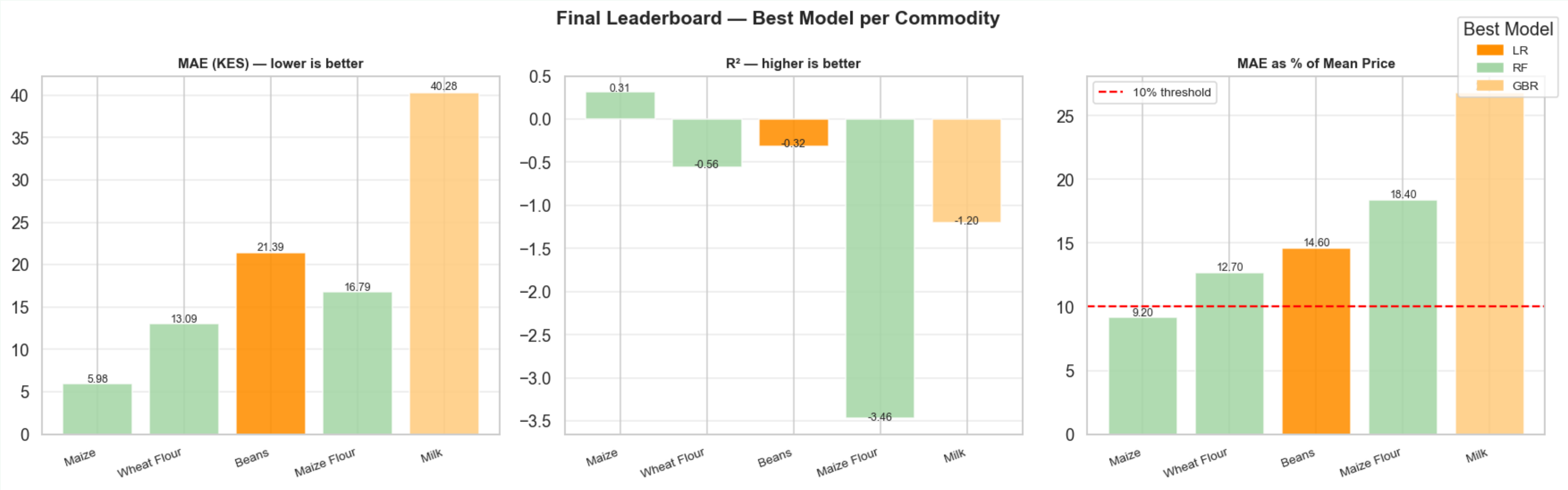
Model slightly under-predicts on average

0.52

R^2 Score

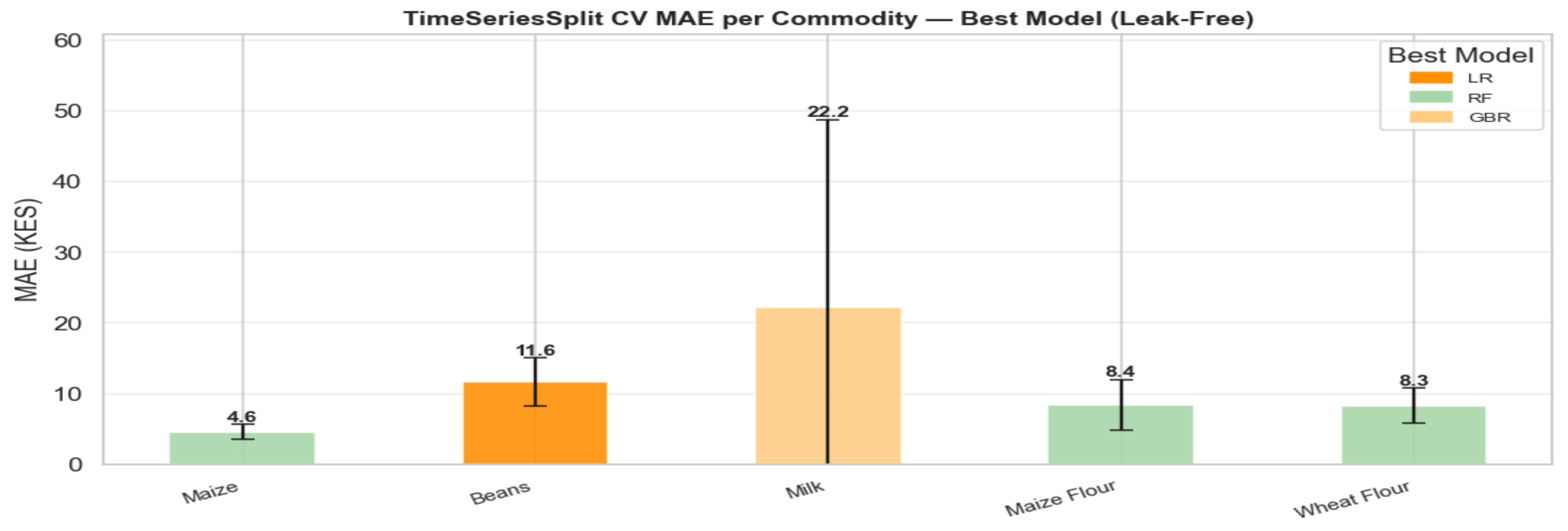
Explains 52% of price variation overall

FINAL RESULTS — BY COMMODITY



Commodity	Best Model	MAE (KES)	% of Price	Note
Maize	RF	5.98	**9.2%**	*Only commodity under 10% error*
Wheat Flour	RF	13.09	**12.7%**	
Beans	LR	21.39	**14.6%**	
Maize Flour	RF	16.79	**18.4%**	
Milk	GBR	40.28	**18.4%+**	*Wide county variation*

CROSS-VALIDATION — HOW RELIABLE ARE THE RESULTS?



- Cross-validation tested the model on 5 different time slices to ensure it wasn't just memorising old data.
- Maize is the most reliably predictable (CV MAE = 4.6 KES ± 1.0). Milk is the hardest (22.2 KES ± 20.2) due to sparse county data.
- Training data: 2010–2023. Test data: 2024–2026 (unseen future prices).

KEY TAKEAWAYS
& NEXT STEPS

- ✅ Maize is the most predictable: 9.2% average error (RF model) — just below the 10% target
- ✅ Past prices are overwhelmingly the strongest signal (96.1% of importance) — markets have strong momentum
- ✅ Fuel price surges (2022–2024) drove food inflation — diesel data meaningfully improves predictions
- ✅ County-level predictions matter: Mandera pays 2.7× more than Kitui for the same beans

Deploy as a mobile-friendly tool for farmers in Uasin Gishu and remote counties

Expand to forecast all 5 commodities with improved models (especially Milk and Beans)